

A Project Report on

Design and development Of 360 degree rotating fire fighting and movement detecting Robot using IOT

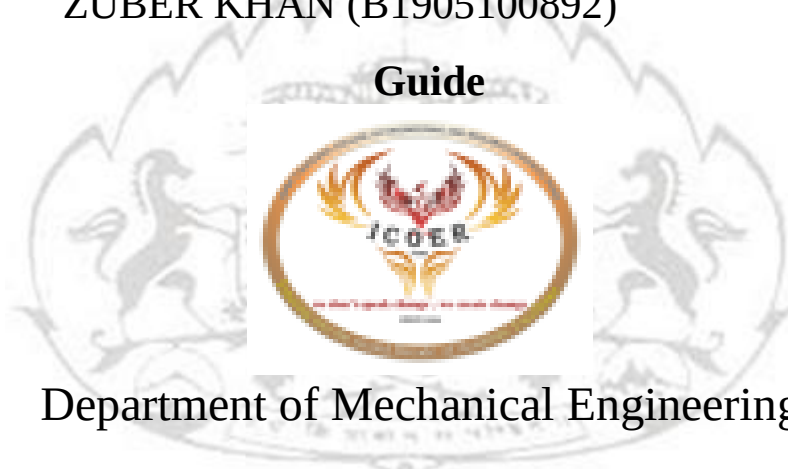
By

PRATIK LAHU DHADE(B1905100810)

AMAN C. JAISWAR (B1905100840)

MAHESH GORSKH PATHARE (B1905100863) ZEESHAN
ZUBER KHAN (B1905100892)

Guide



Department of Mechanical Engineering

JSPM's Imperial College of Engineering and Research,
Wagholi, Pune

[2024-25]

B. E. (Mechanical Engineering) – 2019 Course

Department of Mechanical Engineering



C E R T I F I C A T E

This is to certify that,

- | | |
|---------------------------------|----------------------------|
| 1) <i>PRATIK LAHU DHADE</i> | Exam Seat No.: B1905100810 |
| 2) AMAN C. JAISWAR | Exam Seat No.: B1905100840 |
| 3) <i>MAHESH GORAKH PATHARE</i> | Exam Seat No.: B1905100863 |
| 4) <i>ZEESHAN ZUBER KHAN</i> | Exam Seat No.: B1905100892 |

has successfully completed the Project Work of entitled “**360 degree rotating fire fighting and movement detecting Robot using IOT**” under my supervision, in the partial fulfillment of Bachelor of Engineering (Mechanical Engineering) of Savitribai Phule Pune University.

Date :

Place :

Prof.
Prof.S.P.Joshi

Prof. N.S.Biradar
HOD MECH

Seal

External Examiner

Dr. R.S.Deshpande
Principal, ICOER

B. E. (Mechanical Engineering) – 2019 Course

Savitribai Phule Pune University

ACKNOWLEDGEMENT

It is our proud privilege and duty to acknowledge the kind of help and guidance received from several people in preparation of this report. It would not have been possible to prepare this report in this form without their valuable help, co-operation and guidance.

We express our sincere gratitude to our guide, **Prof.S.P.Joshi Guide**, Department of Mechanical Engineering, ICOER, Wagholi, Pune for guiding us in investigations for this Project and in carrying out experimental work. We hold him in esteem for guidance, encouragement and inspiration received from him.

Our sincere thanks to **Prof. N.S.Biradar**, Head of the Department of Mechanical Engineering, ICOER, for his valuable suggestions and guidance throughout the period of this report.

First and foremost, we wish to record our sincere gratitude to Management of this college and to our beloved Principal, **Dr.R.S.Deshpande**, Principal, JSPM's Imperial College of Engineering and Research, Wagholi, Pune for his constant support and encouragement in preparation of this report and for making available library and laboratory facilities needed to prepare this report.

The Project on “**360 degree rotating fire fighting and movement detecting Robot using IOT**” was very helpful to us in giving the necessary background information and inspiration in choosing this topic for the Project.

1) PRATIK LAHU DHADE	Exam Seat No.: B1905100810
2) AMAN C. JAISWAR	Exam Seat No.: B1905100840
3) MAHESH GORAKH PATHARE	Exam Seat No.: B1905100863
4) ZEESHAN ZUBER KHAN	Exam Seat No.: B1905100892

Chapter No	Description	Page number
1.0	Abstract	01
2.0	Introduction	02
3.0	Motivation	
4.0	Literature Survey	03
5.0	Significance of Topic	17
6.0	Problem Statement	18
7.0	Objectives	19
8.0	Project Flow	
9.0	Methodology	20
9.1	Stage 1	22
9.2	Stage 2	27
9.3	Stage 3	32
10.0	Designing 3D and 2D Model	
11.0	Design and Calculations	
12.0	Components	
13.0	Cost Table	
14.0	Conclusion	
15.0	Future Scope	
16.0	References	

Design and development of 360 degree rotating Fire Fighting Robot Using IOT

Abstract

Fire accidents have become increasingly common in recent years. There are many possibilities to get fire in an Industry or any remote areas. For example, in cotton mills, garments, fuel storages, etc., electric leakage causes huge damage also it's a worst-case scenario, which cause heavy loss not only financially also surroundings got affected. More and more firefighting robots have been

designed to solve the problem of fires. However, the indoor environment is very complex, with high temperatures, thick smoke, more turns, and various burning substances. In this study, a firefighting robot with inspection using fire sensor and camera module and spraying water with the help of the nozzle which is attached to the water pump to suppress the fire functions intended for use in unknown indoor environments was designed. an autonomous inspection process of the on-fire environment, the firefighting robot could identify the flame in real-time, trigger the SMS to house owner or security person , with the help of remote control access system to carry out spraying water with the help of the nozzle which is attached to the water pump to suppress the fire.



Introduction

In most of the countries fire accidents are occurring commonly such as Indian people has suffered and loss many lives because of fire accidents. A fire threat is the third most serious threat to the business continuity and operations. So, to minimize losses and lives resulting from accidents firefighting robots will play the major role. The location of the fire accidents for example: garment factories, gas, petrol pumps and chemical companies etc. This type of accidents results in loss of lives and pollutes the environment. The government and other regulatory has prescribed fire safety

standards and measures. In this project we have made a prototype of remote controlled 360 degree fire protection system which can control the fire without a help of man.



Robot is a machine that looks like a human being and performs various complex tasks. There are many types of robots such as fixed base robot, mobile robot, underwater robot, humanoid robot, space robot and medicine robot etc. In this paper a FIRE FIGHTING ROBOT is proposed. This robot is equipped with a esp 32 module used and feed the signals to the microcontroller in order to trigger the pump which sprinkles water in order to extinguish the fire. This robot is controlled using a mobile phone. This robot implements the concepts of environmental fire sensing, proportional motor control. The motor driver is used for the bidirectional control of the motors equipped in the robot. Every instruction for motion control is given to the robot with the help of ESP.

MOTIVATION

The motivation behind this project is that firefighter's death. Many firefighters are struggling to perform their duty which causes much death while on a mission and the circumstances related to each incident. Firefighters are our heroes and our sense of security in times of trouble. They put themselves on dangerous situations to protect us. At present, the world is moving toward the use of technologies software and hardware

LITERATURE REVIEW:

1. **“Design and Development of Intelligent Fire-fighting Robot Based on STM32”** by Changzhong Wu1,*, Fan Ge2, Guangchao Shang1, Mingpeng Zhao1, Guitao Wang1, Hengshuai Guo1, Liang Wu1

With the rapid growth of economy, fires occur frequently in various dangerous places. According to the investigation, in the ancient tragic fire cases, the number of intelligent mobile fire-fighting equipment is very small, and ordinary fire extinguishers and fire-fighting trains can't put out the fire in time and quickly. As a kind of robot, intelligent fire-fighting robot plays an increasingly important role in fire-fighting and rescue. In this paper, the design of fire robot is studied. Intelligent fire-fighting robot is mainly composed of fire-fighting movement system, fire-fighting water spray system, fire-fighting control system and other components. All parts cooperate with each other and work together to complete the fire-fighting task.

2. “Design and Implementation of an IoT Based Firefighting and Affected Area Monitoring Robot” by Md. Anowar Hossain , Himaddri Shakhar Roy , Md. Fazlul Karim Khondakar , Md. Fazlul Karim Khondakar , Md. Azad Hossain

In this paper, an IoT based fire-fighting and affected area monitoring robot is proposed. This fire-fighting robot can be used as a supplementary to the firefighters in critical situations. To function this robot, a flame sensor, a gas sensor, an IR sensor, and a temperature humidity sensor has been used. The flame sensor is used to detect the fireplace at the same time as the gas sensor informs about the presence of flammable gases, the Passive Infrared Sensor confirms the presence of a human, and the temperature humidity sensor sends statistics about the temperature and humidity of the locality. The robot can run in both manual control system and autonomic control system. This paper discusses the detail and top working condition of a fire-fighting robot and recapitulates an IoT based communication system to monitor the fire affected area using Wi-Fi and also discusses the elaborate functions of each module and the implementation of the system. All the data are sent to the cloud server for further investigation. The proposed firefighting robot has been used for many experiments and proper evaluation has been done based on its performance. It has an excellent performance to extinguish the fire in an emergency situation.

3. “Arduino based firefighting Robot” by S Kirubakaran¹, S P Rithanyaa², S P Thanavarsheni², E Vigneshkumar²

A fire fighter's work entails detecting and extinguishing fires. In this rapidly evolving technological age, the world is gradually moving toward automated systems. Firefighters, on the other hand, are often in danger of losing their lives. The majority of the deaths were caused by toxic gases found in the firefighting environment. As a result, in order to resolve these issues, our system was developed.

4. “Design of Home-use Fire Fighting Robots and Research on Automatic Control System” by Yuchen Ma¹, SiYang Zhang^{1,*}, Yifan Wang¹

Responding quickly to fires, reducing secondary disasters and ensuring the safety of life and property are important issues in firefighting. With the progress of science and technology, firefighting robots have entered people's sight. Based on this, this paper proposes a home-based firefighting robot to reduce fire losses. The home-type firefighting robot is equipped with a walking system and a clamping device. ANSYS software is used to perform finite element analysis on key components to verify the rationality of the structural design. The program mainly uses the PWM module in the K60 chip, and the I/O module is designed. CCD is used for image acquisition and K60 microprocessor. The classical PID control algorithm is still used to make the fire robot achieve the effect of smooth and fast movement. The incremental PID algorithm is used to realize the fast response of the motor

5. “Implementing a Prototype Autonomous Fire Detecting and Firefighting Robot” by Joseph Azeta ^{1*}, Idowu Ayoade ², Cosmas Nwakanma ³ and Timileyin Akande

Fires pose a significant threat to human life and property. An autonomous fire detection and fighting robot has the potential to mitigate the challenges associated with firefighting. This paper presents the implementation of an autonomous fire detection and fighting robot equipped with sensors, algorithms, and actuators to detect, locate, and suppress fires autonomously. The robot's mechanical and electrical components were designed, prototyped, and tested in a controlled environment to evaluate its ability to navigate through obstacles, detect fires, and suppress them autonomously. The results show that the robot effectively detects and suppresses fires, making it a promising solution for firefighting applications. The study contributes to the development of autonomous robots for firefighting, enhancing the safety of firefighters and the effectiveness of firefighting operations.

6. “SELF-DIRECTED FIRE FIGHTING ROBOT USING INTERNET OF THINGS AND MACHINE LEARNING” by Rajeshwarrao Arabelli¹, T.Bernatin²

Now a day, fire accidents in houses, apartments and communities, threatening to the victims and property. As it is a very dangerous job to involve any person like fire fighters during fire accidents, that potentially cause loss of property and human lives due to lack of technology innovation. Hence the firefighting robots are used to rescue the operation instead of humans. In our project, Firefighting robot is used to alert whenever fire accidents are detected and moves in the direction

of flame or smoke to extinguish it. Hence the firefighting robot operation is to rescue victims and stop fire in a house within a little span of time. Thus, it reduces the risk of injury to the victims and also property damage. This device includes various sensors like Proximity Infrared Sensor (PIR), flame sensor, ultrasonic sensor, MQ2 (LPG) sensor, and actuators like Motors and buzzer.



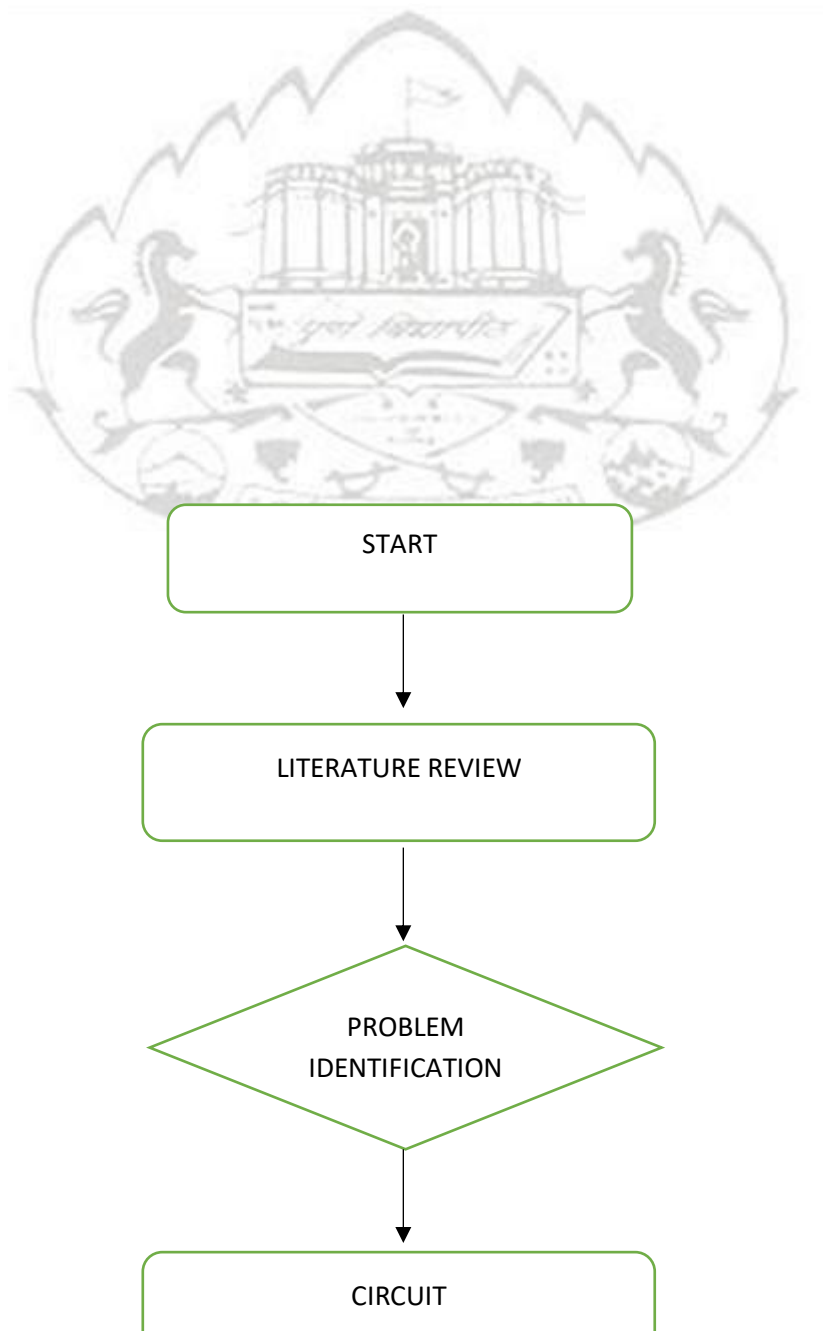
Problem Definition

- A firefighting robot can assist or replace firefighters in carrying out firefighting work, reducing firefighters' labor intensity and improving their firefighting and rescue ability and efficiency.
- Fighting fire within the building requires good thermal protection of the electronic components of the robots. This kind of protection has been considered in the proposed firefighting robot.

Objectives

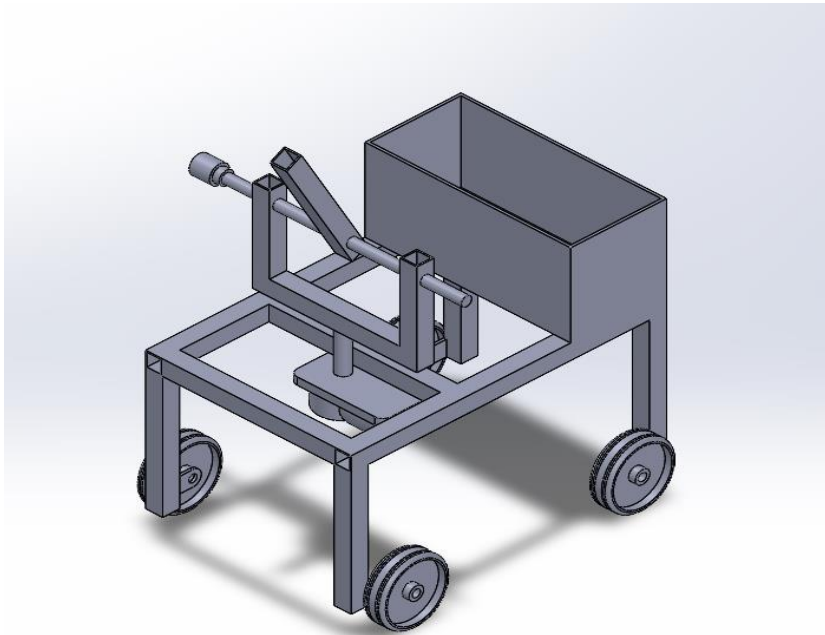
- Develop 360-degree rotating Fire Fighting Robot that can be operate without human involvement
- The firefighting cannon can be remotely controlled for rotation, elevation, and automatic sweeping, providing high flow, long-range, and with the ability to water spray.
- To making a high-definition digital wireless video transmission system for remote real-time video monitoring.
- To adopt frequency-hopping wireless communication technology, enabling long communication distances and strong anti-interference capability.

Project flow:

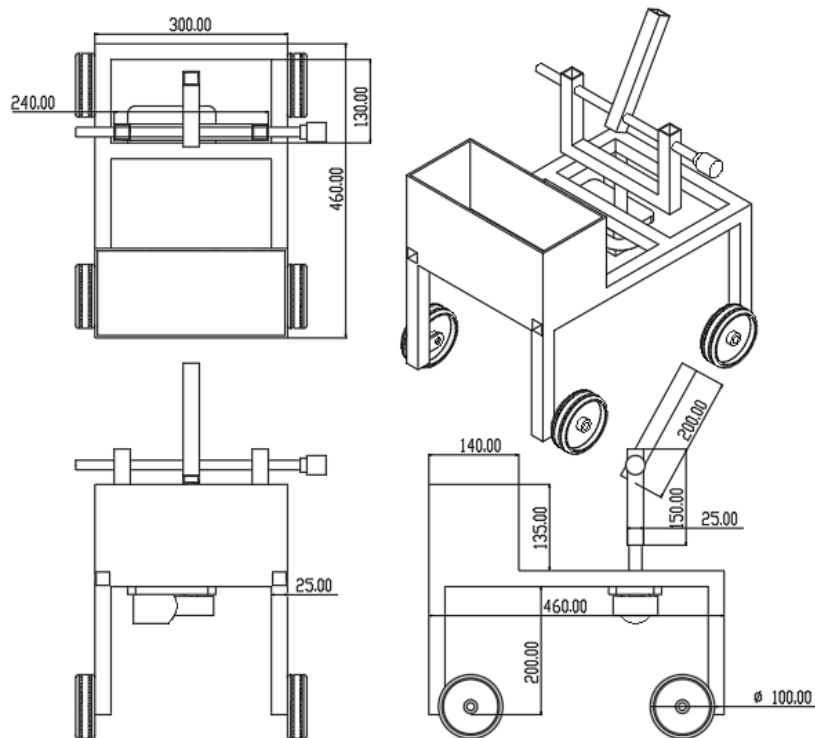


3D CAD MODEL:





2D DRAFTING:



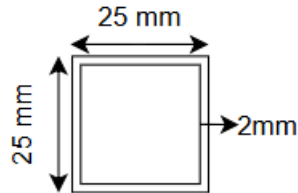
DESIGN AND CALCULATION

Design of frame

Load on frame considered $P = 5 \text{ kg} = 49.05 \text{ N}$

$$y = D/2 = 25/2 = 12.5 \text{ mm}$$

$D = 25 \text{ mm}$ $B = 25 \text{ mm}$ $t = 2 \text{ mm}$ thickness



Length of frame is 460 mm

Moment of inertia in x direction

$$I = 16345.34 \text{ mm}^4$$

$$4Mb = 5640.75 \text{ N-mm}$$

Bending stress of pipe

$$\frac{Mb}{I} = \frac{\sigma b}{y}$$

Where,

- σ is bending stress (Pa or N/m^2)
- M = Bending moment applied to the object ($\text{N}\cdot\text{m}$)
- y = the perpendicular distance between the neutral axis and a specific point on the object's cross-section
- I = the area moment of inertia of the object (m^4)

$$= 4.31 \text{ N/mm}^2$$

Theoretical bending stress

$$\sigma_b(th) = \frac{S_{yt} - 310}{f.s - 1} = 310 \text{ N/mm}$$

Hence design is safe.

CALCULATION FOR DC MOTOR

$$\text{Rpm} = 60$$

$$\text{Voltage} = 12 \text{ v}$$

$$\text{Current} = 0.3 \text{ amp}$$

$$\text{Power} = V \times I$$

$$= 12 \times 0.3$$

$$P = 3.6 \text{ watt}$$

Torque required for motor is

$$P = \frac{2\pi NT}{60}$$

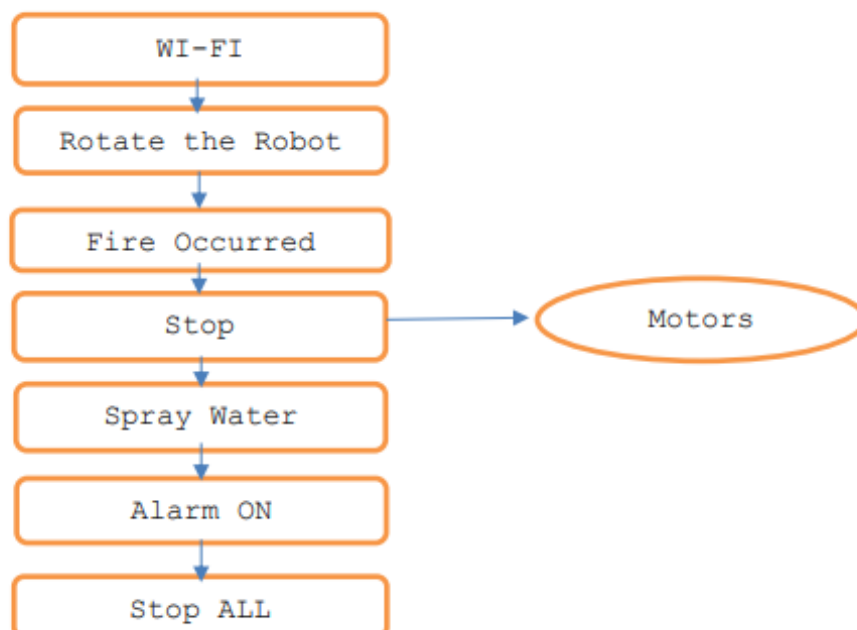
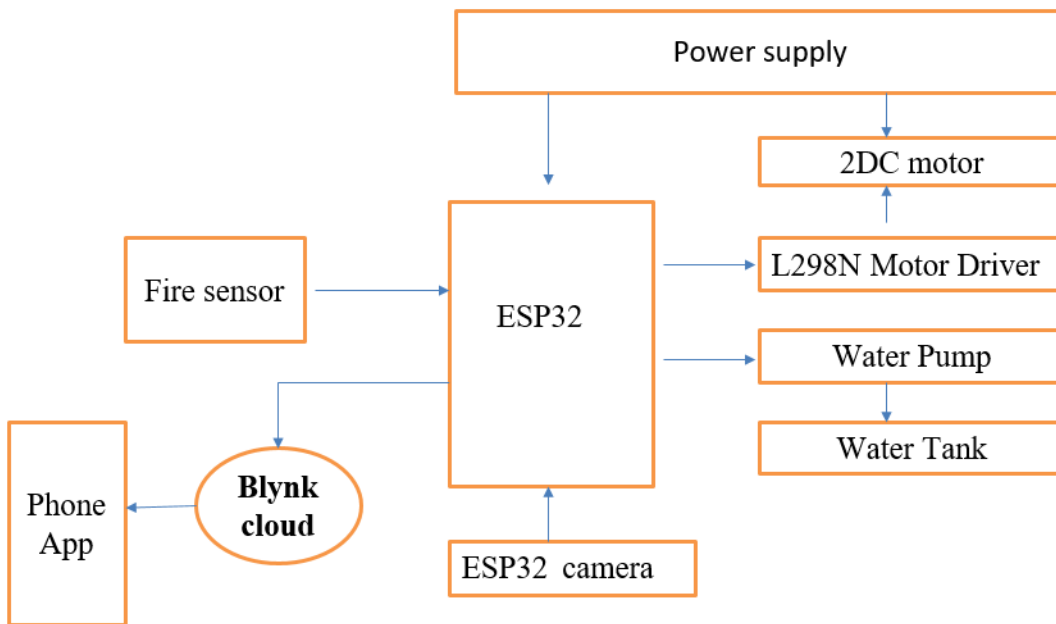
$$3.6 = \frac{2 \times 3.142 \times 60 \times T}{60}$$

$$T = 0.57 \text{ Nm}$$



HARDWARE IMPLEMENTATION

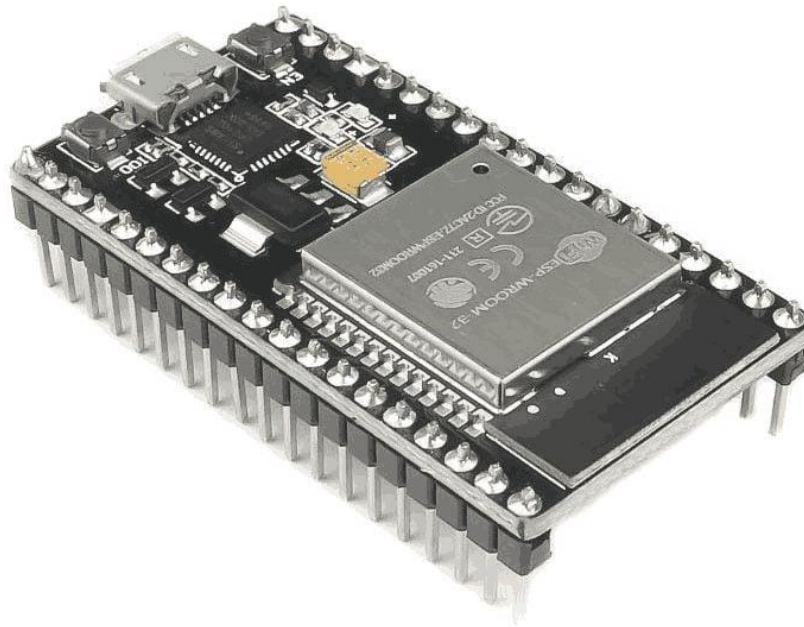
Block Diagram



Working of prototype model:

- ESP 32 is the main processing unit for the functioning of Surveillance robot. The transfer of data from robot to user's mobile after connecting the WIFI to the robot which is present in the ESP32 camera. ESP32 camera has inbuilt WIFI module.
- Firstly, switch on the robot, then the LED will light up which indicates that the robot is in ON condition. Then switch ON the hotspot in the mobile and connect the WIFI module in the ESP32 camera to the robot which is used for transfer of data.
- A user can operate a firefighting robot equipped with a water tank to put out flames in their region using a surveillance firefighting robot with a vision camera.
- The system employs an RF remote control for remote control operation and an RF receive based microcontroller circuit to control the robotic vehicle. Through user commands based on RF, the receiver circuit receives RF signals. Now, the receiver circuit decodes the transmitted data commands.
- The microcontroller is then informed about it. Now that these instructions have been processed, the microcontroller directs the robot to operate the dc motor in the proper direction.
- This enables the operator to control the robot while keeping a safe distance from the flames. Additionally, a wireless camera is put on top of this robot body.
- This camera aids in steering the robot body in the desired direction. This is due to the fact that the wireless camera's captured region can be viewed on a mobile device for reference. As a result, this technology uses the water tank that is attached to the robot body to safely put out fires at a distance.

1. ESP 32



SPECIFICATION OF ESP 32:

- Single or Dual-Core 32-bit LX6 Microprocessor with clock frequency up to 240 MHz
- 520 KB of SRAM, 448 KB of ROM and 16 KB of RTC SRAM.
- Supports 802.11 b/g/n Wi-Fi connectivity with speeds up to 150 Mbps.
- Support for both Classic Bluetooth v4.2 and BLE specifications.
- 34 Programmable GPIOs.
- Up to 18 channels of 12-bit SAR ADC and 2 channels of 8-bit DAC
- Serial Connectivity include 4 x SPI, 2 x I²C, 2 x I²S, 3 x UART.
- Ethernet MAC for physical LAN Communication (requires external PHY).
- 1 Host controller for SD/SDIO/MMC and 1 Slave controller for SDIO/SPI.
- Motor PWM and up to 16-channels of LED PWM.
- Secure Boot and Flash Encryption.
- Cryptographic Hardware Acceleration for AES, Hash (SHA-2), RSA, ECC and RNG.

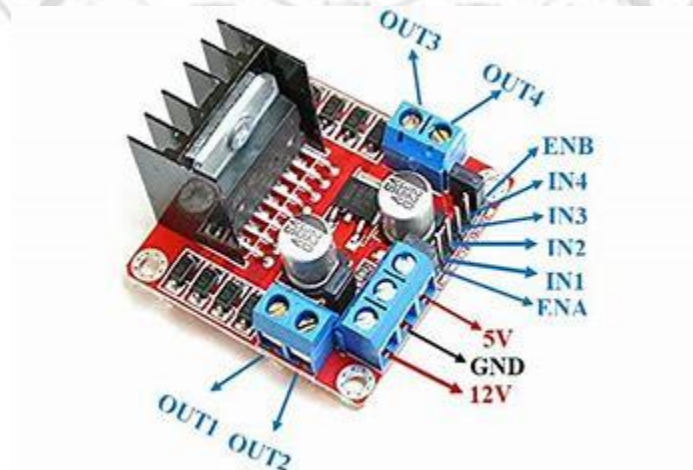
2. DC Motor



Specifications

- Operating Voltage: 12 V DC
- No Load Current: ≤ 220 mA
- No load Speed: 60 RPM (at 12V)
- Full Load Current: ≤ 1300 mA
- Rated Current: ≤ 4800 mA
- Rated Torque: 40 Kg-cm

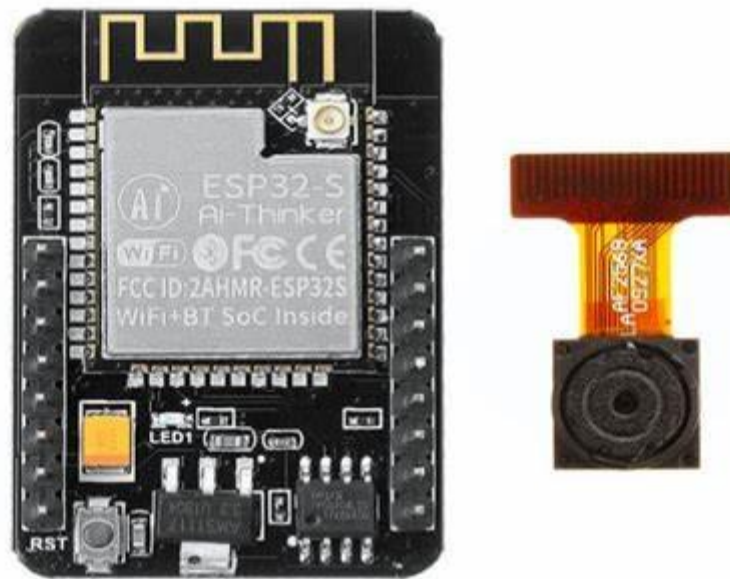
3. L298N Motor Driver



The L298N is a double H bridge drive chip. Its features and specifications include:

- Operating voltage supply up to 46 V
- Total DC up to 4 A
- Saturation voltage is less
- Protection from over temperature
- Power dissipation is 25 W

4. ESP32 camera:



The ESP32 camera specifications are as follows:

- Processor: ESP32-D0WD
- Built-in Flash: 32Mbit
- RAM: Internal 512KB + External 4M PSRAM
- Antenna: Onboard PCB antenna
- WiFi protocol: IEEE 802.11 b/g/n/e/i

5. Flame sensor



Specifications of a flame sensor include:

- Operating voltage: 3.3V to 5V
- Detection range: Few centimetres up to several meters
- Detection angle: Narrow angles such as 60 degrees or 90 degrees
- Response time: Few milliseconds to several seconds
- Sensing wavelength: 760nm to 1100nm



COST TABLE

SR. NO.	COMPONENT	COST
1.	FRAME	3000/-
2.	ESP32	600/-
3.	Dc motor	500/-
4.	Wheel	1000/-
5.	Motor driver	350/-
6.	dc motor	1500/-
7.	Relay	200/-
8.	Shaft	400/-
9	Flame sensors	200/-
10	ESP camera	800/-
11	Water pump and nozzle	600/-

References:

1. Changzhong Wu^{1,*}, Fan Ge², Guangchao Shang¹, Mingpeng Zhao¹, Guitao Wang¹, Hengshuai Guo¹, Liang Wu¹ “Design and Development of Intelligent Fire-fighting Robot Based on STM32” doi:10.1088/1742-6596/1748/6/062019
2. Md. Anowar Hossain , Himaddri Shakhar Roy , Md. Fazlul Karim Khondakar , Md. Fazlul Karim Khondakar , Md. Azad Hossain “Design and Implementation of an IoT Based Firefighting and Affected Area Monitoring Robot” DOI: 10.13140/RG.2.2.19512.37125
3. S Kirubakaran¹, S P Rithanyaa², S P Thanavarsheni², E Vigneshkumar² “Arduino based firefighting Robot” doi:10.1088/1742-6596/1916/1/012204
4. Yuchen Ma¹, SiYang Zhang^{1,*}, Yifan Wang¹ “Design of Home-use Fire Fighting Robots and Research on Automatic Control System” doi:10.1088/1742-6596/2234/1/012014
5. Joseph Azeta ^{1*}, Idowu Ayoade ², Cosmas Nwakanma ³ and Timileyin Akande “Implementing a Prototype Autonomous Fire Detecting and Firefighting Robot” doi:10.20944/preprints202305.2010.v1
6. Rajeshwarrao Arabelli¹, T.Bernatin² “SELF-DIRECTED FIRE FIGHTING ROBOT USING INTERNET OF THINGS AND MACHINE LEARNING” <https://doi.org/10.26782/jmcms.2020.08.00018>

